Cosc 30: Discrete Mathematics

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Problem 1. Two players A and B are playing games with a fair coin with two sides: head and tail. Compute the probability of each of the following item by first coming up with the sample space, the probability of each outcome, and the corresponding event.

1. A and B each throw the coin once, and they get the same side.
2. A throws two coins, hides the result but told B (truthfully) that "there is at least one head". B makes a random guess out of the two and finds a head.
3. A and B take turns to throw the coin until a head appears, starting with A; A ends up getting the first head.

Solution.

1. $1/2$
   - Sample Space $S := \{(i, j) : i, j \in \{H, T\}\}$. The size of S is 4.
   - Probability of each outcome is $(1/2)^2 = 1/4$ per outcome.
   - Event $E := \{(H, H), (T, T)\}$. The size of E is 2.

   Thus the probability that event E happens is $\mathbf{Pr}[E] = \frac{|E|}{|S|} = \frac{1}{2}$.

2. $2/3$
   - Sample Space $S := \{(H, T, k), (H, H, k), (T, H, k) : k \in \{C_1, C_2\}\}$, where C_1, C_2 denote the coins flipped. The size of S is 6.
   - Probability of each outcome is $\left(\frac{1}{3}\right) \cdot \left(\frac{1}{2}\right) = \frac{1}{6}$ per outcome.
   - Event $E := \{(H, H, C_1), (H, H, C_2), (H, T, C_1), (T, H, C_2)\}$. The size of E is 4.

   Thus the probability that event E happens is $\mathbf{Pr}[E] = \frac{|E|}{|S|} = \frac{4}{6} = \frac{2}{3}$.

3. 2/3

- Sample Space $\mathsf{S} := \{\omega\mathsf{H} : \omega \in \{\mathsf{T}\}^*\}$.
- Probability of each outcome is $\left(\frac{1}{2^{k+1}}\right)$, where k is the length of ω. The sum of all possible outcomes is

$$\sum_{k\geq 0}\left(\frac{1}{2^{k+1}}\right) = 1$$

- Event $\mathsf{E} := \{\omega\mathsf{H} : k \pmod 2 \equiv 0, \omega \text{ has even length}\}$.

Thus the probability that event E happens is

$$\mathbf{Pr}[\mathsf{E}] = \sum_{k\geq 0}\left(\frac{1}{2^{2k+1}}\right) = \frac{1}{2}\sum_{k\geq 0}\left(\frac{1}{2^{2k}}\right) = \left(\frac{1}{2}\right)\left(\frac{4}{3}\right) = \frac{2}{3}$$